A Quantity Theory of Money Approach to Cryptocurrency Valuation: Store of Value, Medium of Exchange, and the Economic Moat Signal

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Abstract

We apply the Quantity Theory of Money (QTM) to the cross-sectional valuation of cryptocurrencies and protocol tokens. Treating each digital asset as the currency of a self-contained economy, we decompose market capitalization into a Medium of Exchange (MoE) component, priced by the free-circulating velocity of supply, and a Store of Value (SoV) residual. The key innovation is λ—the fraction of total supply that is locked or staked—which partitions the circulating supply into economically active and inert components. A central result is that the SoV/MoE ratio simplifies to $\lambda/(1-\lambda)$, directly observable from on-chain data without velocity estimation. For native blockchain coins, high λ reflects a convenience yield and seigniorage signal grounded in monetary

search theory. For governance and utility tokens, which lack a medium-of-exchange function, λ itself is an economic moat signal, capturing token holder fundamental conviction in platform growth. Empirically, the signal operates differently across the two asset types. For coins, conviction predicts returns only conditionally: interacted with our Growth-Levelized NVT ratio, a one-standard- deviation-higher staking share earns +0.7% per month among cheap coins and −3.0% among expensive ones (interaction $t = -3.5$), a result that survives a horse race against alternative valuation and technical signals and is robust to seigniorage-yield controls. For governance tokens, the signal is one-dimensional: conviction carries a uniform premium concentrated in the extremes—a top-minus-bottom conviction quintile portfolio earns 1.7% monthly factor-adjusted alpha ($t = 2.2$), strengthening to 2.5% ($t = 3.2$) after controlling for token return reversal, and surviving costs and sub-periods. Mechanism tests attribute the asymmetry to differential absorption of the conviction signal and locate the token premium between protocol categories rather than within them.

1 Introduction

What determines the value of a cryptocurrency? Existing approaches divide into two strands. Academic work prices digital assets in equilibrium models of the transactional and security services a network provides (Pagnotta and Buraschi, 2022; Biais, Bisière, Bouvard, and Casamatta, 2019; Schilling and Uhlig, 2019). Applied and practitioner valuation, by contrast, adapts discounted cash flow (DCF) logic, searching for a dividend-like flow—transaction fees, staking rewards, protocol revenue—that can be capitalized into a present value, typically implemented as fee- or revenue-multiple comparisons. The cash-flow approach, however, mischaracterizes the fundamental nature of digital assets. Cryptocurrencies and protocol tokens exhibit currency-like behavior: each coin or token is the monetary instrument of a self-contained digital economy, and its value is governed by the same forces that determine the value of any currency—the scale of economic activity it enables and the efficiency with which it circulates. The correct framework, we argue, is the Quantity Theory of Money (Fisher, 1911).

The Fisher equation, $MV = PQ$, relates the stock of money (M) and its velocity (V) to the nominal value of economic output (PQ) (Fisher, 1911). Applied to digital assets, a cryptocurrency's market capitalization proxies its money supply, and the economic output it supports—on-chain transaction volume for native coins, protocol throughput for tokens—is its PQ. This is not merely an analogy. Cong, Li, and Wang (2021) develop a formal dynamic model in which token prices aggregate heterogeneous users' transactional demand, with equilibrium prices governed by platform adoption—a formulation directly consistent with the QTM framework we develop here. Sockin and Xiong (2020) similarly model tokens as platform membership instruments, showing that the token price reflects the present value of the convenience yield accruing to participants: the non-pecuniary benefit of network membership, including access to platform services, network liquidity, and future transactional capability.

Before proceeding, we define the two components of value our framework decomposes.

The *Medium of Exchange* (MoE) value of a digital asset is the value attributable to its current transactional function: the economic activity it enables today, priced by the velocity with which it circulates. The *Store of Value* (SoV) value is the premium agents are willing to pay today by forgoing current exchange utility—the transactions, goods, services, or alternative investments they could obtain by spending the asset now—in anticipation of future value appreciation. This appreciation is expected to arise from network growth, expanding user adoption, increasing utility, and innovation. An agent holding Ethereum rather than spending it is implicitly paying an intertemporal price: she sacrifices today's exchange utility for her conviction that the network's future worth exceeds today's use value. The SoV component is the aggregate market valuation of this forward-looking conviction.

The QTM framework requires a critical extension to operationalize this decomposition. Not all of a cryptocurrency's circulating supply is economically active. A substantial and growing fraction is locked in staking contracts, pledged as governance collateral, or held in long-term custody with near-zero velocity. These holdings are economically inert from a transactional standpoint: they contribute to the money supply but perform no current exchange function. Conflating active and inactive supply distorts velocity estimates and, consequently, misprices the economic content of market capitalization. We address this by introducing λ, defined as the proportion of the total supply that is locked or staked—including tokens committed to network validation, governance contracts, and other long-term holding functions. The formal derivations are provided in Appendix A. The key result is that the SoV/MoE ratio simplifies to

$$\frac{\mathrm{MC}_{\mathrm{SoV}}}{\mathrm{MC}_{\mathrm{MoE}}} = \frac{\lambda}{1-\lambda}, \tag{1}$$

a monotone, directly observable function of λ alone, requiring no velocity estimation.

Why does λ matter beyond its role in the decomposition? For native blockchain coins—Ethereum, Solana, and their peers—high λ is grounded in three complementary theoretical traditions. First, *monetary search theory* (Kiyotaki and Wright, 1993; Lagos and Wright, 2005) establishes that money emerges as a coordination equilibrium: an asset becomes valuable because rational agents expect others to hold and accept it. High λ is the blockchain-observable, publicly verifiable signal of this coordination commitment. Second, the *convenience yield* (Sockin and Xiong, 2020; Krishnamurthy and Vissing-Jorgensen, 2012) is the non-pecuniary benefit agents derive from holding a monetary asset: access to network security, platform liquidity, and future transactional capability. Just as holders of U.S. Treasury securities accept lower yields in exchange for liquidity services (Nagel, 2016), holders of dominant blockchain coins accept lower required returns because network membership has value beyond current transactions. Crucially, the SoV premium is the present value of expected *future* convenience yields: agents who lock coins today are paying for anticipated growth in the network's membership value. Third, the *seigniorage* channel (Cong and He, 2024) anchors this signal in rational optimization: in proof-of-stake networks, newly minted tokens distributed to stakers constitute monetary seigniorage, and the staking decision reveals belief that this income plus anticipated price appreciation exceeds the opportunity cost of locking capital. These three channels are complementary: coordination theory explains why high λ is self-reinforcing, convenience yield provides the holding motive, and seigniorage grounds the rational optimization.

Governance and utility tokens present a related but distinct case, and the distinction matters for how the framework is applied. A governance token such as UNI is the monetary instrument of the Uniswap protocol economy, but it has no medium-of-exchange function: the token does not need to change hands to facilitate Uniswap trades, so there is no MoE component to separate from an SoV premium. With the MoE denominator absent, the SoV/MoE ratio is undefined for tokens, and we use λ itself to convey the same conviction content directly. The protocol's economic output is still well defined—decentralized exchange volume, liquidity deployed, fees generated—and whether protocol revenue currently accrues to token holders is immaterial, just as it is immaterial whether U.S. GDP flows to dollar holders. This framing aligns with Cong, Li, and Wang (2021), who price tokens on the present value of future platform utility and adoption, and Schilling and Uhlig (2019), who model the value of digital currencies as functions of the economic activity they support. For governance tokens, λ carries a different signal. Staking a governance token into a DAO contract renders it illiquid: staked tokens cannot be sold while locked. An agent who voluntarily incurs this illiquidity cost reveals a preference for future appreciation over present liquidity—a choice grounded in assessments of network effects, utility expansion, competitive positioning, and innovation trajectory. Drawing on Hirschman (1970), holders who exercise *voice* (governance participation) rather than *exit* (selling) reveal a preference for the protocol's future that passive buyers do not. Edmans and Manso (2011) formalize this in a corporate setting: blockholders who engage rather than exit create governance value precisely because engagement is costly and therefore credible. Applied to tokens, active governance participation is a costly signal (Spence, 1973) of fundamental conviction in platform growth, making high participation rates an economic moat indicator.

We operationalize these ideas in two metrics per asset type. For coins, the *SoV/MoE ratio*—equal to $\lambda/(1-\lambda)$—measures the balance between forward-looking conviction and current transactional utility, and is our primary return predictor; for tokens, λ itself plays this role, as described above. The second metric normalizes market capitalization by fundamental economic activity, and its denominator likewise adapts to the asset type. For coins, the *Growth-Levelized NVT ratio* normalizes market capitalization by the present value of the network's expected transaction stream PQ (constructed analogously to a PMT-based discounted cash flow, as detailed in Section 3). For tokens, protocol throughput is measured by Total Value Locked rather than transaction volume, so the corresponding metric is the *Growth-Levelized NV/TVL ratio*: market capitalization over the growth-adjusted present value of TVL, built with the same discounting machinery as the coin metric. Crucially, neither valuation ratio carries its own independent theoretical prediction for returns. Rather, each serves as a conditioning variable: the conviction signal is most valuable as a return predictor when the market has not yet priced it in—that is, when the valuation ratio is low. Assets that combine high conviction with cheap valuation (Stars) offer the best expected returns; assets that combine low conviction with expensive valuation (Avoid) offer the worst.

Our results support the framework in a sharper, conditional form. For coins, the conditioning mechanism is the result: conviction alone does not predict returns, but its interaction with valuation is strongly negative ($t = -3.5$) — a one-standard- deviation-higher staking share earns +0.7% per month among cheap coins and −3.0% among expensive ones, a result that survives a horse race against raw NVT, stock-to-flow, and technical signals, and is robust to seigniorage-yield controls. For governance tokens, the signal proves one-dimensional: conviction carries a premium of roughly 0.7% per month per standard deviation regardless of valuation, concentrated in the sort extremes — a long–short portfolio of top-minus-bottom conviction quintiles earns 1.7% per month in factor-adjusted alpha ($t = 2.2$), rises to 2.5% ($t = 3.2$) when controlling for the strong token reversal effect, and survives transaction costs and sub-period splits. The coarse two-dimensional quadrant portfolio itself does not earn significant alpha at our universe breadth, a power-limited null we report in full. Mechanism tests attribute the cross-track asymmetry to differences in how the market absorbs the conviction signal, with the token premium residing largely between protocol categories rather than within them. The horse race also disciplines the cash-flow approach on its own terms: the practitioner price-to-fees multiple is the one alternative valuation ratio with cross-sectional predictive power for tokens, yet its own DCF refinement destroys that power — discounting adds value where the fundamental is a throughput flow, and subtracts it where the fundamental is already a cash flow.

Related Literature. A growing body of work attempts to value digital assets, and our paper connects to several strands. On equilibrium cryptocurrency valuation, Pagnotta and Buraschi (2022) model Bitcoin value as a function of user adoption and network security;

Biais, Bisière, Bouvard, and Casamatta (2019) study equilibrium selection in blockchain protocols; Schilling and Uhlig (2019) develop a general equilibrium model of Bitcoin as a competing currency; and Easley, O'Hara, and Basu (2019) analyze how transaction fees connect mining incentives to market value. Cong, Li, and Wang (2021) and Sockin and Xiong (2020) are the closest to our approach in modeling token prices through adoption dynamics and convenience yields respectively; we complement them by providing an empirically tractable QTM decomposition and a testable investment signal derived from on-chain observables. On crypto risk factors and return predictability, Liu and Tsyvinski (2021) identify network, momentum, and investor attention factors; Liu, Tsyvinski, and Wu (2022) develop a three-factor model for cryptocurrency returns; and Makarov and Schoar (2020) document price dislocations across exchanges. We contribute to this literature the first on-chain supply-side signal—$\lambda/(1-\lambda)$—grounded in monetary theory rather than return-based factor construction. On the monetary economics of digital assets, Kiyotaki and Wright (1993), Lagos and Wright (2005), and Schilling and Uhlig (2019) provide search-theoretic and general equilibrium foundations; Cong and He (2024) analyzes proof-of-stake tokenomics; and Jiang, Krishnamurthy, and Lustig (2024) draw the analogy between dollar exorbitant privilege and dominant-network premia. The convenience yield literature (Krishnamurthy and Vissing-Jorgensen, 2012; Nagel, 2016) motivates the monetary premium in $\mathrm{MC}_{\mathrm{SoV}}$. On token issuance and governance, Howell, Niessner, and Yermack (2020) study the determinants of ICO success; Chod and Lyandres (2021) model ICO design under information asymmetry; and Edmans and Manso (2011) and Hirschman (1970) provide the governance theory linking holder participation to firm and protocol value. Our paper is the first, to our knowledge, to derive a directly observable valuation decomposition from QTM, to distinguish the signal content of λ for coins versus governance tokens, and to demonstrate that the interaction of conviction and valuation predicts cross-sectional cryptocurrency returns.

The remainder of the paper is organized as follows. Section 2 develops the theoretical frameworks for coins (SoV/MoE) and tokens (governance moat) and derives the testable propositions. Section 3 defines and constructs the three key variables: the conviction index λ, the Growth-Levelized NVT ratio for coins, and the Growth-Levelized NV/TVL ratio for tokens. Section 4 describes the data sources, sample construction, and summary statistics. Section 5 presents the cross-sectional return predictability and portfolio results. Section 6 reports robustness and mechanism tests. Section 7 collects all findings in a single summary. Section 8 concludes.

2 Theoretical Framework

This section develops the theoretical argument in three steps. Section 2.1 establishes the coin framework: QTM applied to native blockchain coins yields a SoV/MoE decomposition in which the locking ratio λ is the sufficient statistic. Section 2.2 develops the token framework: governance tokens lack a medium-of-exchange function, so λ is reinterpreted as a governance moat signal grounded in costly signaling theory. Section 2.3 introduces the common valuation-conditioning mechanism that applies across both tracks. Each subsection closes with the empirical propositions we test.

2.1 The Coin Framework: SoV/MoE and the Staking Ratio

The QTM identity applied to a native blockchain coin treats market capitalization as money supply: $\mathrm{MC} \cdot V = PQ$, where V is the velocity of supply in transactional use and PQ is nominal on-chain output. The critical observation is that supply is heterogeneous in velocity. A fraction λ of total circulating supply is locked in proof-of-stake validation contracts, staked to earn seigniorage, or otherwise immobilized with near-zero velocity. The complementary fraction $(1-\lambda)$ is freely circulating and economically active. Only the active fraction supports transactions; the locked fraction is held as a store of value and does not circulate. Partitioning supply accordingly, the appendix shows that market capitalization decomposes as

$$\mathrm{MC} = \underbrace{\frac{PQ}{V_{\text{active}}}}_{\mathrm{MC}_{\text{MoE}}} + \underbrace{\frac{PQ}{V_{\text{active}}} \cdot \frac{\lambda}{1-\lambda}}_{\mathrm{MC}_{\text{SoV}}},$$

and therefore the SoV/MoE ratio satisfies

$$\frac{\mathrm{MC}_{\text{SoV}}}{\mathrm{MC}_{\text{MoE}}} = \frac{\lambda}{1-\lambda}.$$

This is an accounting identity given λ, not yet a behavioral prediction.

The behavioral prediction follows from modeling the individual staking decision. Consider a continuum of agents indexed by $i \in [0, 1]$, each choosing at date t whether to stake one unit of the native coin or hold it freely circulating. Staking yields a publicly observed current reward $b_t > 0$—seigniorage paid to validators in a PoS network (Cong and He, 2024)—plus a private convenience yield $c_{i,t+1} = \theta_{t+1} + \varepsilon_i$, where θ_{t+1} is the network's true future growth trajectory (unobserved at t) and ε_i is mean-zero idiosyncratic signal noise. Staking also imposes a private liquidity cost φ_i (i.i.d., independent of θ_{t+1} and ε_i). Agent i stakes if and only if

$$b_t + \theta_{t+1} + \varepsilon_i \geq \varphi_i \quad \Longleftrightarrow \quad \eta_i \leq b_t + \theta_{t+1}, \qquad \eta_i \equiv \varphi_i - \varepsilon_i.$$

By the law of large numbers, the aggregate staking ratio converges almost surely to

$$\lambda_t = H(b_t + \theta_{t+1}),$$

where H is the CDF of η_i. Idiosyncratic cost and signal noise wash out in the cross-section; what survives is a deterministic, strictly increasing function of θ_{t+1}. Because b_t is publicly observed, λ_t is a publicly verifiable signal of the holder base's aggregate private conviction about the network's future growth. Extracting θ_{t+1} from λ_t, however, requires knowing H and b_t with precision—not common knowledge for most market participants. As in Grossman and Stiglitz (1980), the signal is not fully impounded into price at t. As θ_{t+1} materializes and becomes public over $t+1$, price converges toward the value anticipated by high-λ stakers, generating positive return predictability.

For coins, the empirical counterpart of the conviction signal is therefore the SoV/MoE ratio itself, $\lambda/(1-\lambda)$, constructed from the observed staking share. The formal hypotheses, common to both asset classes, are stated in Section 2.4.

2.2 The Token Framework: Governance Conviction as Economic Moat

For governance and DeFi protocol tokens, the SoV/MoE framework is inapplicable: these tokens are not media of exchange. A holder of UNI does not need to spend UNI to use Uniswap; holding governance tokens does not reduce the protocol's transactional capacity. The token has no meaningful velocity in the QTM sense, and the SoV/MoE decomposition carries no economic content. What λ signals for tokens is something different: *governance conviction*—the fraction of supply whose holders have voluntarily incurred a cost to demonstrate fundamental belief in the protocol's future.

The formal model is a straightforward extension of the coin model. The same continuum of agents now hold a governance token at date t and choose whether to lock it into a governance contract (vote-escrow, delegation, or protocol staking) or hold it freely. Locking carries a private cost $\varphi_i = \ell + \kappa_i$, decomposed into a common illiquidity cost $\ell \geq 0$ (positive for vote-escrow protocols such as Curve's veCRV or Balancer's veBAL, zero for simple delegation under ERC20Votes) and an idiosyncratic engagement cost $\kappa_i \geq 0$, the monitoring and time cost of exercising governance voice rather than exit (Hirschman, 1970; Edmans and Manso, 2011). The current locking benefit $b_t \geq 0$ may include a fee or revenue share for protocols with on-chain distribution; for most governance tokens $b_t = 0$ and the incentive to lock is purely forward-looking. The threshold condition is identical to the coin case: $\eta_i = \varphi_i - \varepsilon_i \leq b_t + \theta_{t+1}$, yielding $\lambda_t = H(b_t + \theta_{t+1})$.

The key difference is interpretation. For coins, $b_t > 0$ is paid in newly minted seigniorage, so staking is partly a rational income decision. For tokens with $b_t = 0$, locking is a purely conviction-revealing action: the holder incurs $\ell + \kappa_i$ with no current compensation, revealing that their private belief about θ_{t+1} is strong enough to make the commitment worthwhile. This is a Spence (1973) costly-signaling equilibrium: only holders with sufficiently high conviction rationally pay the cost. The resulting aggregate λ_t is therefore a cleaner signal of fundamental conviction for governance tokens than for coins, because the seigniorage confound is absent. The partial-revelation logic applies identically: the signal in λ_t is not freely extractable from public data, so it is not instantly priced, generating the same return predictability.

For tokens, the empirical counterpart of the conviction signal is the composite locking index λ itself, since the SoV/MoE transformation is undefined without a medium-of-exchange denominator. The mechanism is the same in both classes—a costly, publicly observable action aggregates private beliefs about θ_{t+1}, and costly extraction prevents instantaneous pricing—so the hypotheses below apply to coins and tokens alike, with only the measurement of conviction and valuation differing by class.

2.3 The Valuation Conditioning Signal

The locking model establishes that λ carries information not yet fully in price. What they do not characterize is how much of the signal remains unpriced at a given date. Let $\delta_t \in [0, 1]$ denote the fraction of the conviction signal in λ_t already absorbed into the current price, so that $p_t = p_t^{\text{fund}}(1 + \delta_t)$ where p_t^{fund} reflects current economic activity. The expected return as the signal is validated over $t + 1$ is proportional to the unabsorbed portion $\lambda_t(1 - \delta_t)$, not to λ_t alone. The marginal predictive power of λ_t is therefore decreasing in δ_t, and δ_t is increasing in the current market capitalization relative to fundamental activity.

For coins, the relevant fundamental is discounted expected on-chain volume: δ_t is increasing in $\text{NVT\_GL}_t = \text{MC}_t/PQ_t^*$, where PQ_t^* is the growth-adjusted present value of expected transaction activity (formally defined in Section 3.2). For tokens, the relevant fundamental is the growth-adjusted present value of Total Value Locked: δ_t is increasing in $\mathrm{NV/TVL\_GL}_t = \mathrm{MC}_t/\mathrm{TVL}_t^*$, where TVL_t^* is constructed with the same discounting machinery (Section 3.3). In both cases, a high ratio indicates that current market capitalization already embeds the conviction signal, and the incremental return predictability of λ_t is low; a low ratio indicates the signal remains unabsorbed.

The valuation ratio carries no independent return prediction in this framework. A low NVT_GL alone—without a high λ—merely describes a coin with limited current throughput relative to market cap, not necessarily a mispricing; similarly for NV/TVL_GL. The conditioning role is asymmetric: valuation identifies the *timing* of conviction signal absorption, not a separate source of alpha. One theoretical qualification matters for interpreting the tests: the conditioning prediction has empirical content only where the absorption fraction δ_t *varies* in the cross-section. If $\delta_{j,t}$ is approximately uniform within a class, expected returns are proportional to $\lambda_{j,t}$ alone and the interaction is degenerate—the conviction signal becomes one-dimensional. Whether δ_t varies, and whether the chosen valuation ratio measures that variation, are empirical questions to which we return in Section 6.2.

2.4 Testable Hypotheses

The locking model and the conditioning argument yield three hypotheses. They are stated once, for both asset classes: the economic mechanism—costly, observable commitment aggregating private beliefs, absorbed into prices only gradually—is common, and the classes differ only in how conviction and valuation are measured (Section 3): the SoV/MoE ratio and NVT_GL for coins, the locking index λ and NV/TVL_GL for tokens.

Hypothesis 1 (Conviction). *Assets with higher conviction in month t earn higher returns over month $t+1$, controlling for size, momentum, prior-month return, and market beta.*

Hypothesis 2 (Conditioning). *The predictability in H1 is concentrated among as-*

sets that are cheap relative to fundamental economic activity: the conviction $\times$ valuation interaction is negative, and the conviction premium is larger below the class-month median valuation ratio than above it.

Hypothesis 3 (Quadrant portfolio). *A monthly-rebalanced portfolio long Star assets (above-median conviction, below-median valuation) and short Avoid assets (below-median conviction, above-median valuation) earns positive factor-adjusted alpha against market, size, and momentum factors, within each asset class.*

H3 is the integrating test: it asks whether the joint two-dimensional signal has incremental predictive power beyond either dimension alone. Because the measurement of both dimensions differs across classes, all tests are run within class, and any divergence in verdicts across classes is informative about where the common mechanism operates—a question the results indeed raise, and to which the mechanism analysis of Section 6.2 responds.

3 Variable Construction

The theoretical framework in Section 2 requires three empirical proxies: the conviction index λ, common to both asset tracks; the Growth-Levelized NVT ratio for the coin track; and the Growth-Levelized NV/TVL ratio for the token track. This section defines each variable and describes its construction. Data sources and sample formation are covered in Section 4.

3.1 The Conviction Index λ

The theoretical λ_t is the aggregate locking or commitment ratio of an asset's holder base at date t. Empirically, the conviction-revealing actions observable on-chain vary by asset type and protocol design. We therefore construct λ as a composite index over three conceptually motivated channels, and aggregate across available channels using cross-sectional z-scores.

Channel 1: Staking. For proof-of-stake layer-one coins, the staking channel is the fraction of circulating supply bonded as validator or delegator stake at month-end. This is the direct empirical counterpart of the locking action in Section 2.1: it simultaneously earns seigniorage ($b_t > 0$) and constitutes a costly, publicly visible commitment to the network's future. Staking rates are sourced from chain-native archive data for each coin.[1] The staking channel covers 48 assets and 2,222 asset-months.

Channel 2: Holding. The holding channel measures the fraction of circulating supply that has not transacted on-chain in the preceding six months (HODL-6m). This proxy extends conviction measurement to any EVM-compatible asset without a formal staking mechanism, mapping to the inert supply fraction λ_t in the model: a holder who has not transacted for six months has revealed a preference for holding over liquidity. HODL-6m is computed by replaying on-chain Transfer event logs in chronological order under a FIFO attribution rule, assigning each unit of supply to its most recent transfer date.[2] This channel spans 408 assets and 11,654 asset-months, providing the broadest coverage in the conviction panel.

Channel 3: Governance. Two governance sub-channels capture active participation in protocol decision-making. The delegation sub-channel records the fraction of circulating supply delegated to a voting representative under an ERC20Votes contract, observable from on-chain Delegate event logs.[3] The voting sub-channel records proposal participation rates from Snapshot off-chain governance records, cross-referenced with on-chain ERC20Votes logs

[1]For Polkadot (DOT) and Kusama (KSM), bonded supply is retrieved from the Subscan archive API. For Cosmos-ecosystem chains (CRO, KAVA, OSMO), we query the Cosmos SDK Staking module endpoint at block heights corresponding to month-end timestamps, with timestamps identified via binary search on the CometBFT RPC interface. For EVM-based staking contracts (BNB, MATIC, TRX, and others), staked supply is derived from on-chain staking event logs via the Etherscan multi-chain archive API. Each data source is validated by confirming that reported values vary across block heights, ensuring the archive reflects historical rather than current-only state.

[2]Transfer events are retrieved from the Etherscan Pro V2 multi-chain archive, covering Ethereum mainnet and twelve additional EVM chains. Two data quality filters are applied: (i) monthly on-chain flow exceeding 100× reported circulating supply is classified as a minting, bridging, or aggregator artifact and excluded; (ii) asset-months in which HODL-6m exceeds 80% persistently are flagged for robustness checks, consistent with possible custodial cold-wallet concentration rather than individual holder conviction. Self-transfers (sender equals recipient) are excluded from the event replay.

[3]Data sourced from `DelegateVotesChanged` event logs via the Etherscan Pro V2 API, aggregated to month-end totals.

where available. Both sub-channels identify holders who have incurred the engagement costs κ_i of Section 2.2—the costlier the action, the more selected and informative the resulting signal. The delegation and voting sub-channels cover 26 and 53 assets, respectively, concentrated in the DeFi governance token sample.

Composite Index. For each asset-month, each available channel is standardized to a z-score over all asset-months with non-missing observations for that channel. The composite λ_z is the equal-weight average of standardized channel scores available for that asset-month. The result has approximately zero cross-sectional mean and is defined whenever at least one channel is observed. Equal weighting imposes no prior on the relative information content of the channels; channel-specific predictive power is examined in robustness tests. The full conviction panel covers 467 assets and 13,626 asset-months, of which 11,901 are single-channel, 1,371 two-channel, and 354 three-or-more-channel observations.

3.2 Growth-Levelized NVT

For the coin regression track, the valuation anchor is the Growth-Levelized Network Value to Transactions ratio (NVT_GL), which scales market capitalization by the growth-adjusted, risk-discounted present value of expected on-chain throughput:

$$\mathrm{NVT\_GL}_{j,t} = \frac{\mathrm{MC}_{j,t}}{PQ^*_{j,t}}, \tag{2}$$

$$PQ^*_{j,t} = \frac{\displaystyle\sum_{s=1}^{n} \frac{PQ_{0,j,t}(1+g_{j,t})^s}{(1+r_{e,j,t})^s} + \frac{PQ_{0,j,t}(1+g_{j,t})^n(1+g_\infty)}{(r_{e,j,t}-g_\infty)(1+r_{e,j,t})^n}}{\dfrac{1-(1+r_{e,j,t})^{-n}}{r_{e,j,t}}}, \tag{3}$$

where $PQ_{0,j,t}$ is the trailing twelve-month annualized on-chain transfer volume for asset j at month t; $g_{j,t}$ is the trailing three-year compound annual growth rate of PQ_0, capped in $[-50\%, 200\%]$; $n = 10$ years is the explicit-growth projection horizon; $g_\infty = 3\%$ is the terminal perpetuity growth rate; and $r_{e,j,t} = r_f + \hat{\beta}_{j,t} \cdot \text{MRP}$ is the asset-specific discount rate, with $r_f = 4\%$, MRP $= 30\%$, and $\hat{\beta}_{j,t}$ the trailing 36-month beta against the BTC price index, floored so that $r_{e,j,t} \geq g_\infty + 2\%$ to ensure well-defined terminal values.[4]

NVT_GL below one signals that market capitalization is below the growth-adjusted present value of expected on-chain activity. NVT_GL substantially above one indicates that current market capitalization already embeds forward growth not yet visible in throughput. The growth adjustment corrects a systematic bias in raw NVT: fast-growing assets may appear expensive on a raw NVT basis but inexpensive once expected growth is discounted.

3.3 Growth-Levelized NV/TVL

For the token regression track, the valuation anchor is the Growth-Levelized Network Value to Total Value Locked ratio, constructed with the same discounted-present-value machinery as NVT_GL. Total Value Locked—the market value of assets deposited in a protocol's smart contracts—proxies for the scale of economic activity the protocol intermediates, and is the token analog of on-chain transaction volume in the QTM framework. Because TVL is a stock rather than a flow, the base is a trailing average level rather than a trailing sum:

$$\text{NV/TVL\_GL}_{j,t} = \frac{\text{MC}_{j,t}}{\text{TVL}^*_{j,t}}, \tag{4}$$

$$\text{TVL}^*_{j,t} = \frac{\displaystyle\sum_{s=1}^{n} \frac{\text{TVL}_{0,j,t}(1+g_{j,t})^s}{(1+r_{e,j,t})^s} + \frac{\text{TVL}_{0,j,t}(1+g_{j,t})^n(1+g_\infty)}{(r_{e,j,t}-g_\infty)(1+r_{e,j,t})^n}}{\dfrac{1-(1+r_{e,j,t})^{-n}}{r_{e,j,t}}}, \tag{5}$$

where $\text{TVL}_{0,j,t}$ is the trailing twelve-month average of month-end TVL for protocol j (requiring at least six monthly observations); $g_{j,t}$ is the trailing three-year compound annual

[4] The denominator of (3) is the standard annuity factor normalizing the finite-horizon present value of future PQ to an equivalent annualized flow, making NVT_GL interpretable as a price-to-normalized-activity ratio analogous to a price-to-normalized-earnings multiple in equity analysis. The MRP of 30% reflects the elevated required return on cryptocurrency risk; $\hat{\beta}_{j,t}$ is retained in the panel so alternative discount rate assumptions can be applied directly in robustness tests.

growth rate of TVL_0, with two- and one-year fallbacks and the same $[-50\%, 200\%]$ cap; and all remaining parameters—n, g_∞, r_f, MRP, and the beta construction—are identical to those in (3), so the coin and token tracks share a single set of valuation assumptions. Where a protocol is deployed across multiple chains, TVL is the sum of all-chain TVL.

The growth adjustment serves the same purpose as in the coin track: a protocol whose TVL is expanding rapidly may appear expensive on a raw NV/TVL basis but inexpensive once expected growth is priced in, and conversely a protocol with collapsing TVL may appear deceptively cheap on the raw ratio. A low NV/TVL_GL indicates that market capitalization is modest relative to the growth-adjusted present value of the protocol's economic footprint; a high NV/TVL_GL indicates the reverse.

4 Data

4.1 Data Sources

Price, market capitalization, and circulating supply are drawn from CoinMarketCap at daily granularity, snapped to end-of-calendar-month observations. CoinMarketCap provides the broadest historical coverage of digital asset market data and is the standard reference for universe construction in the cryptocurrency finance literature (Liu, Tsyvinski, and Wu, 2022; Makarov and Schoar, 2020).

On-chain Transfer event logs, staking event logs, and ERC20Votes Delegate event logs are retrieved from the Etherscan Pro V2 multi-chain API, which provides archive access to EVM-compatible chains including Ethereum mainnet, BNB Chain, Polygon, Avalanche, Arbitrum, Optimism, and nine additional networks. Non-EVM staking data follows chain-native sources: Subscan archive API for Polkadot and Kusama; Cosmos SDK Staking module endpoints for Cosmos-ecosystem chains. Off-chain governance participation is sourced from the Snapshot API, cross-referenced with on-chain ERC20Votes logs.

On-chain transaction volume for the NVT_GL construction is assembled from three sources in priority order. DeFiLlama chain-level DEX volume[5] serves as the primary source for chains with active decentralized exchange infrastructure (1,340 asset-months). BitInfoCharts sent-in-USD settlement value[6] is used for PoW coins and pre-DEX periods, capturing all on-chain transfers rather than DEX activity alone (705 asset-months). Blockchain.com estimated transaction value[7] supplements Bitcoin and legacy chain observations with pre-DEX-era coverage (118 asset-months). The remaining 363 asset-months use DeFiLlama protocol-level volume for assets whose primary activity flows through specific protocols. Each series is validated against the chain's own reported aggregate, with the reconstructed monthly level required to lie within 5% of the chain total. The resulting NVT_GL panel covers 67 assets and 2,526 valid asset-months from August 2016 through May 2026.

Total Value Locked is sourced from DeFiLlama at daily granularity, snapped to month-end; multi-chain protocols use the sum of all-chain TVL. The raw TVL panel covers 162 assets and 8,066 asset-months from September 2017 through May 2026. Applying the growth-levelization of (5)—which requires a trailing twelve-month TVL average and at least one year of prior history for the growth estimate—yields an NV/TVL_GL panel of 111 assets and 3,125 asset-months from January 2021 through May 2026.

4.2 Sample Construction

The starting universe is 1,939 digital assets drawn from the CoinMarketCap historical database, observed at end-of-calendar-month from August 2015 through May 2026, yielding 156,838 asset-month observations.[8] Table 1 traces the path from this universe to the two regression samples.

Each asset is classified as a *coin* (native settlement asset of a public blockchain), a *token* (smart-contract-issued asset on a host chain), or *other* (stablecoins, wrapped and

[5] `https://defillama.com`, accessed through May 2026.

[6] `https://bitinfocharts.com`, accessed through May 2026.

[7] `https://blockchain.com`, accessed through May 2026.

[8] All assets with at least one non-zero price observation in CoinMarketCap are included. Market capitalization is computed as reported price times circulating supply. Continuous coverage is not required; assets may enter and exit the panel as they gain or lose listing.

bridged representations, and infrastructure assets whose price is pegged or mechanically derived); the 858 other-class assets are excluded outright, since neither valuation framework applies to them, leaving 633 coins and 448 tokens. The coin track requires a staking channel (Section 2.1): removing 323 proof-of-work coins, 24 with unverifiable consensus, 256 proof-of-stake coins whose historical staking state is unrecoverable from public archives, and 6 without overlapping on-chain volume leaves the *coin regression sample* of 24 assets and 718 asset-months (December 2020 through May 2026; median NVT_GL 0.017, median market capitalization $820 million). The token track requires TVL: removing 299 tokens without DeFiLlama coverage, 3 without any conviction channel, and 45 with TVL histories too short for the growth-levelized denominator leaves the *token regression sample* of 101 assets and 2,771 asset-months (January 2021 through May 2026; median NV/TVL_GL 2.29, median TVL $42.2 million, median market capitalization $98 million).[9] The full membership of both samples is listed in Appendix Tables 16 and 17.

Survivorship deserves explicit comment. The universe itself is constructed from point-in-time monthly snapshots and therefore includes assets that were subsequently delisted or became inactive; no continuous-coverage requirement is imposed, an asset's final observed month contributes its last available return, and returns are never computed across carried-forward observations. In this respect the universe is free of classic survivorship bias. The regression samples, however, condition on data coverage — recoverable staking archives for coins, DeFiLlama TVL reporting for tokens — and coverage correlates with scale and longevity. The samples therefore speak to the covered population of larger, more established assets, a selection that is visible in the median market capitalizations of Table 2; it is a coverage tilt rather than a conditioning-on-survival bias, since coverage providers retain the histories of failed protocols and dead assets remain in the panel through their final months.

[Table 1 about here.]

[9] The growth-levelization requirement shortens the token sample relative to a raw NV/TVL alternative (136 assets, 4,627 asset-months from March 2020); the raw sample is retained for robustness comparisons.

The combined regression panel spans 125 unique assets and 3,489 asset-months. At the end of the sample in May 2026, the panel accounts for \$451 billion in market capitalization—19.0% of the \$2.38 trillion CoinMarketCap universe—consistent with the concentration of on-chain governance and staking infrastructure among the larger assets in the market. Channel 2 (holding) applies only to EVM-compatible assets; PoW coins with UTXO-based ledger accounting require a separate pipeline based on UTXO age analysis and are left to future work.

4.3 Summary Statistics

Table 2 reports the distributions of the main variables in the two regression samples; data-panel coverage counts are in Appendix Table 13. Three features of the data deserve comment before the results.

First, the conviction measures display wide, economically meaningful variation. The median proof-of-stake coin has 38.6% of its circulating supply staked, but the interquartile range runs from 21.8% to 57.0%: locking behavior differs enormously across networks, which is precisely the cross-sectional variation the tests exploit. In SoV/MoE terms, the median coin-month ratio of 0.63 says that store-of-value demand is worth roughly two-thirds of transactional demand for the typical coin, while at the 90th percentile (ratio 2.5) the monetary premium is two and a half times the transactional component — the decomposition is not a theoretical curiosity but a first-order feature of observed valuations. The token conviction index, a cross-sectional z-score by construction, is close to centered (median -0.18) with fat enough tails (p10 to p90 spanning -0.86 to $+1.03$) to make quintile extremes meaningfully different populations.

Second, the valuation ratios are extremely dispersed and right-skewed, in an instructive way. Median NVT_GL of 0.02 against a 90th percentile of 2.65 says the typical PoS coin trades far below the discounted value of its throughput while a minority trade well above it; median NV/TVL_GL of 2.29 against a 90th percentile of 281 makes the same point for tokens. By contrast, the price-to-fees multiple is comparatively well-behaved (median 12.1, interquartile range 4.3–33.8) — fee-anchored valuation produces multiples in a recognizable equity-like range, one reason it functions as a level signal where the TVL-based ratio does not. These tails motivate the log transforms and 1%/99% winsorization applied throughout; no test in the paper relies on the raw ratio scale.

Third, the two samples are more alike in risk than in size. Coins are an order of magnitude larger (median market capitalization \$823 million versus \$98 million), but both classes carry monthly return volatility near 30% and near-identical market betas (median 1.12 and 1.16 against BTC) — the return-generating environment is common, the scale is not. Median forward returns are negative in both panels (-4.4% and -7.6% per month), reflecting a sample window dominated by the post-2021 contraction; within-month cross-sectional tests are immune to this level, but it disciplines any reading of long-only portfolio legs.

[Table 2 about here.]

5 Results

This section reports the tests of Hypotheses 1–3 in the order developed in Section 2, followed by a conviction-extremes portfolio analysis whose origin we disclose below, and a horse race against alternative valuation and technical signals. Throughout, dependent variables are one-month-ahead returns winsorized at the 1%/99% monthly cross-section; all regressors are standardized within class-month, so coefficients are per one standard deviation; panel specifications include month fixed effects with standard errors two-way clustered by asset and month. In all tables, *, **, and *** denote significance at the 10%, 5%, and 1% levels, with t-statistics in parentheses.

5.1 Conviction and the Cross-Section of Returns (H1)

Table 3 reports the coin-track specification ladder. Unconditionally, the staking-based conviction signal does not predict next-month coin returns: the coefficient is +0.55% per SD alone ($t = 0.69$) and indistinguishable from zero with controls. Hypothesis 1 is rejected for coins at the one-month horizon. The most consistently significant coin-level characteristic in this sample is size: larger coins earned *higher* returns (+2.3% to +2.7% per SD, $t \approx 3.3$–5.2), the opposite of the equity size premium and consistent with the mega-cap-led 2023–26 recovery.

[Table 3 about here.]

The token track differs. Table 4 shows a positive governance conviction premium of +0.5% to +0.8% per month per SD, but its significance is confined to two specifications: the single-regressor column ($t = 1.97$) and the Fama–MacBeth cross-sections ($t = 1.77$ and 2.19). Once controls enter, with or without coarse protocol-sector fixed effects, the panel slope sits at $t \approx 1.5$. The divergence between the two estimators is itself informative. The Fama–MacBeth specifications include the same controls and are estimated on the full token cross-section each month; they differ from the pooled panel in weighting every month equally rather than by observation count, and in computing standard errors from the time series of monthly coefficients rather than by two-way clustering. Because the token conviction premium is substantially larger in the early, thin-cross-section months (the pre-2023 slope is +2.9% per SD), equal month-weighting raises the Fama–MacBeth estimate, while clustering by asset — which absorbs the strong within-asset persistence of λ — widens the pooled standard errors. The linear slope's significance is thus sensitive to estimator weighting because the premium is not uniform over time; the portfolio-extremes result below does not share this sensitivity. H1 in the token sample thus receives qualified support at best as a linear slope — a limitation that the portfolio analysis of Section 5.4 resolves in an informative direction: the conviction information is real but concentrated in the extremes of the distribution, where a linear coefficient underweights it. Two further qualifications are developed below: sector fixed effects *attenuate* the slope (part of the premium is a between-category effect), and the slope does not survive a joint multivariate race against technical signals (Section 5.5).

[Table 4 about here.]

5.2 Valuation Conditioning (H2)

The central conditional prediction receives sharply asymmetric support. For coins, the conviction–valuation interaction in Table 3 column (4) is -1.69% per SD×SD ($t = -3.54$); the split-sample difference test (column 7) is -4.20% ($t = -3.99$). A one-SD-higher staking share earns $+0.70\%$ per month among cheap (below-median NVT_GL) coins and -3.0% among expensive coins. The pattern strengthens with horizon: at six months the interaction reaches -9.0% ($t = -2.99$) and the unconditional conviction slope turns marginally positive ($+6.1\%$, $t = 1.90$), consistent with slow price discovery. Notably, the negative slope among expensive coins exceeds what pure non-absorption ($\delta_t \leq 1$) can deliver: conviction that is both visible and already richly priced predicts *negative* returns, suggesting the market overshoots on observable staking commitment. The economically supported statement is thus stronger than H2 as written: for coins, valuation does not merely time the conviction premium; the high-conviction–high-valuation corner is actively avoided.

For tokens, H2 is rejected: the interaction is near zero with the wrong sign in every specification ($+0.34\%$, $t = 0.79$ baseline; split difference $t = 0.65$), and Section 6 shows this is not a measurement artifact of the growth-levelized denominator. Token conviction predicts returns *uniformly* across the valuation distribution — the signal is one-dimensional. The mechanism tests of Section 6.2 discriminate among explanations; anticipating the punchline, the token premium proves to be a portfolio-extremes and between-sector phenomenon, not an attention-friction story.

5.3 The Quadrant Portfolio (H3)

Hypothesis 3 is not supported. Table 5 reports Stars-minus-Avoid portfolios: no variant earns significant factor-adjusted alpha against the monthly LTW-analog factors, point estimates flip sign between equal- and value-weighting for coins, and the token estimates are carried by pre-2023 months (post-2023 EW alpha -0.79%/month, $t = -0.73$). The failure is robust to the breadth guard but poorly powered at every feasible setting. The pre-registered construction requires four names per leg from a median cross-section of eleven: 45 of 65 coin months fail the guard, leaving twenty non-contiguous months.[10] And the median split discards the continuous variation that the regressions exploit: a signal of $+0.7\%$ per month per SD, present in half the sample, cannot surface as significant alpha in 20–49 months at this breadth. Transaction costs are not the binding problem (turnover is 0.10–0.28 per leg-month; a 50 bps haircut subtracts $\sim$9 bps). The single-dimension median-split comparators fail equally (all $|t| < 1.8$). We report H3 as a power-limited null: the cross-sectional evidence of Tables 3–4 does not aggregate into a coarse tradable portfolio at this universe breadth.

[Table 5 about here.]

5.4 Conviction Extremes: The Token Quintile Portfolio

Disclosure. Based on the observation that H2 is rejected for tokens while the conviction level receives support, we conducted an exploratory analysis of conviction-sorted portfolios and found that sharper sorts strengthened the signal. The full battery reported here — sort granularities, weighting schemes, costs, sub-periods, sector variants, and spanning tests — was then specified ex ante and run once. We describe the finding as exploratory-then-confirmed, not pre-registered.

[10] Relaxing the guard trades diversification for span and does not change the verdict: at a minimum of three names per leg the coin Stars-minus-Avoid alpha is -2.7% per month ($t = -1.36$, 33 months); at two names per leg, -1.5% ($t = -0.79$, 46 months). The point estimate is negative at every threshold; two-name legs are effectively pair trades dominated by idiosyncratic volatility, which is why the four-name guard was pre-registered.

Table 6 reports conviction-only token long-shorts across sort granularities. The quintile EW portfolio earns +2.28% per month raw and +1.71% factor-adjusted ($t = 2.18$, annualized Sharpe 0.86), survives 50 bps per-side costs (+1.53%, $t = 1.95$) and the post-2023 sub-period (+1.48%, $t = 1.79$). The progression across granularities is non-monotonic and must be read as such: median and tercile splits are dead ($t = 0.38$, 0.37), the quintile works, and the decile dies ($t = 0.02$ at ∼5.8 names per leg). The quintile sits where signal purity and leg breadth (∼10.6 names) balance; we do not claim a monotone sharpening result.

[Table 6 about here.]

Two structural facts sharpen the interpretation. First, the location of the premium within the token universe requires care. Coarse-sector fixed effects attenuate the regression slope (Table 4), and the pooled sector-neutralized quintile is insignificant ($t = 0.39$) — but the latter construction demeans conviction within categories that hold a median of one to eight tokens per month, injecting noise rather than removing sector tilts. The informative test is the one category deep enough to support a within-sector quintile: DEX, with 39 tokens and a median of 25 per month, where the conviction long-short earns +3.4% per month ($t = 2.04$) on legs of fewer than five names. The premium is therefore present within the deepest category at the granularity where the signal lives; the remaining categories are too thin to test at that granularity, and how much of the aggregate premium is between rather than within categories cannot be settled at current breadth. The value-weighted variant adds a complementary fact: its point estimate exceeds the equal-weighted one (+3.1%, $t = 1.83$), so the premium is not a microcap artifact, though the VW series is noisier (ten-name legs concentrate in one or two large tokens) and does not survive the spanning battery on its own; the equal-weighted portfolio remains the result. Second, the spanning tests in Table 7 show the premium is emphatically not a repackaged technical signal. Against the strong token reversal (a winner-minus-loser quintile portfolio loses −4.7% per month in this sample, $t = -3.02$), controlling for reversal exposure *raises* the conviction alpha to +2.53% ($t =$

3.18), because the high-conviction leg tilts toward recent winners; against the momentum-family-plus- size battery the alpha is +2.34% ($t = 2.60$); and against the completed twelve-competitor battery — adding moving-average distance, volatility, idiosyncratic volatility, Amihud illiquidity, and skewness, the last plausible spanners — it is +1.74% ($t = 2.45$), rising to +2.17% ($t = 2.82$) when the two fee-based comparators join. No competitor portfolio earns positive alpha on the conviction quintile in the reverse direction.

[Table 7 about here.]

5.5 Horse Race Against Alternative Signals

We race the conviction signals against a completed comparator battery: raw NVT and NV/TVL, stock-to-flow, supply growth, 52-week high, moving-average distance, volatility, idiosyncratic volatility, Amihud illiquidity, return skewness, the momentum family, and — for tokens — the two implementable versions of the cash-flow approach named in the introduction: the practitioner price-to-fees multiple (P/F, market capitalization over trailing twelve-month protocol fees) and a revenue DCF ratio (market capitalization over the growth-levelized present value of protocol revenue, built with the machinery of Section 3.2).[11]

The joint specifications in Table 8 carry the section's central question: holding every alternative signal fixed, does the conviction signal still predict returns? A coefficient that survives this test contains information no combination of existing signals spans — the regression counterpart of improving on the known predictor set. Signal-by-signal results are in Appendix Table 14; three learnings follow.

[Table 8 about here.]

[11] Protocol fees and revenue are from DeFiLlama, covering 55 and 51 of the 101 sample tokens respectively, with histories mostly beginning in 2021; fee-based tests speak for the covered half of the sample. Revenue — the share of fees accruing to the protocol or its holders — is scarcer and later than fees in every dimension of coverage, itself a fact about disclosure practice in DeFi. Monthly RSI, MACD, and Bollinger-band signals are deterministic transformations of the momentum, moving-average-distance, and volatility measures already in the battery, and daily-native signals (the MAX effect, true 14-day RSI) require daily price histories unavailable at free-tier depth; we omit the former as redundant and note the latter as a data limitation. The Amihud measure uses month-end snapshot volume rather than aggregated monthly volume and is accordingly a noisy illiquidity proxy.

First, *the coin cross-section is nearly information-free at the characteristic level, and the one robust exception is the paper's conditional term.* No signal — including ours — predicts coin returns as an unconditional level over the full sample (Appendix Table 14; the only significant single is Amihud illiquidity). But with all twelve comparators held fixed, the conviction $\times$ valuation interaction remains -2.35% per SD$\times$SD ($t = -2.68$; -1.88%, $t = -3.17$ post-2024). Our signal is not competing with better predictors and losing; it is the only reliable structure in coin returns, and it survives precisely because it captures conditional information that no level characteristic contains. The Amihud result — illiquid coins underperform by 2.8% per month per SD jointly ($t = -4.49$) — is a complementary liquidity discount, not a rival: both terms coexist in the same regression. (Post-2024 singles such as the 52-week-high ratio are regime-specific and do not survive the full sample.)

Second, *the token conviction slope is spanned in the panel, and saying so plainly bounds the claim.* Jointly, the token linear slope attenuates to $+0.11\%$ ($t = 0.19$); in the panel, idiosyncratic volatility and an anomalous positive momentum (12–2) term carry the significance. Combined with Section 5.4, the accurate statement is that token conviction is a portfolio-extremes phenomenon — unspanned in portfolio space by any of the twelve competitors (Table 7) — rather than a uniform cross-sectional slope. The two facts are not contradictory: a signal concentrated in the tails is exactly the kind a linear panel coefficient underweights and a quintile sort recovers.

Third, *the cash-flow foil earns a partial vindication with an instructive twist.* The practitioner price-to-fees multiple is the one alternative valuation ratio with genuine token-level power: -1.03% per SD alone ($t = -2.50$) and -0.90% ($t = -2.32$) with the entire battery held fixed — the only valuation ratio in either track to survive a joint race. Yet its own DCF refinement destroys it (the growth-levelized fee and revenue versions are dead, $t = -0.42$ and -0.68), and it has no portfolio translation (Table 7: own alpha $t = -0.10$). Growth-levelization adds value where the fundamental is a throughput flow (coins, where it is load-bearing for the interaction) and subtracts it where the fundamental is already a cash flow. The fee multiple neither spans nor revives conviction: the conviction slope on the fee-covered subsample is unchanged by the P/F column. Finally, the scarcity narratives popular in practitioner work find no support: stock-to-flow is insignificant everywhere, and supply growth matters only for coins post-2024. A descriptive Metcalfe panel (seven majors only) confirms the well-known Bitcoin price-to-Metcalfe mean reversion ($t = -3.69$ in-sample) but cannot enter the cross-sectional race for coverage reasons.

6 Robustness and Mechanisms

6.1 Measurement Robustness

Table 9 re-estimates the interaction specification under alternative measurement choices. The coin interaction is insensitive to the discount rate assumption (MRP 20%/40%: $t = -3.55/-3.54$) and *strengthens* when the 47 coin-months using the fallback conviction measure are excluded (-1.81%, $t = -4.67$). Two caveats attach, reported with equal prominence. Under raw (non-growth-levelized) NVT the interaction shrinks to a third of its magnitude (-0.60%, $t = -1.90$): the growth adjustment is load-bearing for the coin result. And excluding the 40% of coin-months where the throughput growth cap binds preserves the magnitude (-1.63%) but not significance ($t = -0.97$, $n = 406$, 24 clusters). The token interaction is dead in every variant — raw NV/TVL ($t = 0.05$), cap-excluded ($t = -0.54$), custodial-flag-excluded ($t = 0.55$) — establishing that the token-track H2 null is not an artifact of the growth-levelization machinery.

As a final, model-free check on the token conditioning null, we replace the valuation ratio with normalized market capitalization itself as the absorption proxy: if the conviction signal is priced into visible assets first, the premium should concentrate among small tokens regardless of how fundamentals are measured. It does not. The conviction $\times$ size interaction is $+0.0003$ ($t = 0.07$), and the size-median split runs in the wrong direction: the conviction slope is $+0.14\%$ per month among small tokens ($t = 0.32$) and $+1.01\%$ among large tokens

($t = 1.67$).

The final and cleanest measurement candidate is a fee-anchored denominator: protocol fees and revenue are earned in external assets and are not mechanically linked to the token's own price the way TVL is, so if the conditioning null were a denominator artifact, it should revive here. Table 15 reports the full interaction specification under each alternative valuation ratio. It does not revive. Under both the practitioner fee multiple and the revenue-DCF ratio, the interaction carries the predicted sign for the first time — and the split-sample slopes order correctly (cheap above expensive) — but at $t = -0.45$ and $t = -0.31$ it is a null, and the coverage-matched baselines show no denominator effect to credit.

The quadrant portfolio (H3) was also re-formed under the alternative valuation measures, with construction identical to Table 5. It remains null. With ln P/F as the valuation axis, the token Stars-minus-Avoid portfolio earns +1.67% per month in factor-adjusted alpha — the best quadrant point estimate in the paper, with an annualized Sharpe of 0.82 — but at $t = 1.35$ over 46 months it is insignificant, and it vanishes post-2023 (+0.09%, $t = 0.07$). With the revenue-DCF ratio, the quadrant is negative (−1.89%, $t = -1.24$) on only 31 months, with 33 months lost to the breadth guard on the revenue-covered subsample. The coverage-matched NV/TVL_GL quadrant on the same fee-covered subsample is equally null ($t = 0.68$). Across all three valuation measures, the two-dimensional portfolio adds nothing over the one-dimensional conviction quintile of Table 6 — fully consistent with the regression evidence that the second dimension carries no conditioning information for tokens. The token H2 null thus survives six distinct conditioning variables — the growth-levelized ratio, the raw ratio, sector-demeaned valuation, turnover, market capitalization, and fee/revenue anchoring — and cannot be attributed to the choice of denominator: the conditioning failure is a fact about tokens, not about measurement. One external-validity note attaches: on the revenue-covered subsample (33 larger, revenue-reporting protocols, effectively 2023 onward), the token conviction slope itself turns negative (−1.24%, $t = -2.01$), a composition fact echoing the fragility of the linear slope — in contrast to the sort extremes — documented in Section 5.5. For coins, the same exercise yields a new fact: conviction $\times$ size is *positive* ($+0.0329$, $t = 2.21$), with the conviction premium negative among small coins (-3.2% per month, $t = -1.89$) — consistent with high advertised staking ratios on small chains marking yield traps rather than informed conviction — and the valuation interaction survives alongside the size interaction, attenuated (-0.0134, $t = -1.96$; the two conditioners are nearly orthogonal, $\rho = -0.10$). Valuation and size condition the coin conviction signal through separate channels.

[Table 9 about here.]

6.2 Mechanism Discrimination (M1–M4)

The results present a puzzle the theory did not anticipate: the conditioning mechanism operates for coins but not for tokens. This section follows the natural learning process — observe the asymmetry, postulate mechanisms that could produce it, and test each. We postulated four. *M1 (attention-dependent absorption)*: coin staking ratios are widely published while token governance locks are read by few, so the token absorption fraction δ may be uniformly low; if so, conditioning for tokens should operate on attention proxies, with the premium concentrated in small, low-turnover tokens. *M2 (denominator endogeneity)*: TVL is denominated in assets correlated with the token itself, so NV/TVL variation may reflect business-model differences rather than mispricing; if so, the interaction should revive when valuation is demeaned within protocol category. *M3 (growth-adjustment measurement)*: the TVL growth cap binds in a large share of token-months; if the null is a measurement artifact, it should revive under the raw ratio or excluding capped months. *M4 (seigniorage confound, a threat to the coin result)*: high λ in a richly valued coin may reflect yield-chasing rather than informed conviction; the coin interaction must survive staking-yield controls or the conditioning result is a b_t phenomenon rather than a δ phenomenon. Table 10 reports the verdicts.

[Table 10 about here.]

The pattern is informative in both directions. On the defensive side, M4 clears the principal threat to the coin result: controlling for the staking yield and its interaction with λ, the conviction–valuation interaction is unchanged (-1.88%, $t = -4.39$) — the coin conditioning result is a δ phenomenon, not a seigniorage-crowding artifact. M2 and M3 jointly establish that the token H2 null is real rather than mismeasured. But M1, the mechanism we considered most plausible ex ante, fails in the direction that matters: the token conviction premium is not concentrated among small, low-turnover, attention-starved tokens — the turnover interaction has the wrong sign ($t = 1.21$), and the tercile gradient tilts toward *high*-turnover names. The structural evidence of Section 5.4 disciplines any replacement story: the premium is present within the deepest protocol category (the DEX-only quintile earns $+3.4\%$ per month, $t = 2.04$), attenuated by category fixed effects, and invisible to coarse sorts — a pattern consistent with gradual repricing of governance value operating both within and across categories, concentrated in the extremes of the conviction distribution, but not with individual-token attention neglect of the M1 form. We offer this as an interpretation consistent with the evidence rather than a tested mechanism; the categories outside DEX are too thin to test at the granularity where the signal lives, and distinguishing gradual repricing from unmodeled category risk premia is a task for longer samples.

6.3 Heterogeneity

A pre-specified heterogeneity battery (all cells run, all reported) yields largely null results, three of which discipline the theory. First, the vote-escrow prediction of Section 2.2 — costlier locks, purer signals — is not supported: classifying all 101 tokens by governance mechanism, the conviction slope among vote-escrow tokens ($+0.16\%$, $t = 0.21$) is *smaller* than among plain-governance tokens ($+0.47\%$, $t = 1.21$), with an insignificant interaction in the pooled specification ($t = -0.27$). Second, fee-share status makes no difference ($t = -0.05$), though as a between-token comparison this rules out only large static differences in the b_t comparative static. Third, conviction *changes* ($\Delta\lambda$ over one and three months) carry no information beyond levels in either track, as regressors or as sorts. Size and turnover tercile splits are flat. Regime splits show the coin interaction is concentrated post-2023 (-2.05%, $t = -4.32$; the pre-2023 coin panel has only 74 observations and is uninformative) and mildly stronger in bull months (-2.41%, $t = -1.83$) than bear months (-0.88%, $t = -0.98$); the token conviction slope is stable across regimes.

Finally, decomposing the token conviction index into its constituent channels identifies where the information resides. Table 11 re-estimates the baseline specification and the quintile long-short using each raw channel in place of the composite. The holding-duration channel — the fraction of supply that has not moved in six months — carries essentially all of the signal: its panel slope ($+0.55\%$ per SD, $t = 1.50$ across 88 tokens) and quintile alpha ($+1.48\%$ per month, $t = 1.71$) approach the composite's, while the staking, delegation, and voting channels are individually uninformative on their sparse coverage (9–32 tokens each) and too thin to sort. The composite's modest edge over holding alone ($+1.71\%$ versus $+1.48\%$ alpha) is the increment the governance channels contribute where observed. The economic reading is pointed: what predicts token returns is *revealed patience* — supply that demonstrably does not move — rather than observable governance participation, consistent with the earlier finding that voting-participation weighting adds nothing and with the vote-escrow null above.

[Table 11 about here.]

7 Summary of Findings

Because the empirical analysis spans three hypotheses, an extension, and an extensive horse race, Table 12 collects every verdict in one place before the concluding discussion.

[Table 12 about here.]

Three sentences organize the table. On the hypotheses: the two-dimensional framework survives for coins in its conditional form — conviction matters where valuation has not absorbed it, and visibly rich conviction is punished — while for tokens only the signal's level survives, and no conditioning variable modulates it. On the extension: the token premium is real but lives in the extremes of the conviction distribution and between protocol categories, robust to every technical and fee-based competitor we can construct, with its exploratory origin disclosed. On the horse race: no established valuation or technical signal displaces either result, and the one competitor with genuine token-level power — the practitioner fee multiple — is itself a level effect that dies under its own DCF refinement, reinforcing the paper's broader point that discounting machinery helps only where the fundamental is a throughput flow rather than a cash flow.

8 Conclusion

We derive a directly observable valuation decomposition from the Quantity Theory of Money—the SoV/MoE ratio equals $\lambda/(1-\lambda)$—and show that the on-chain conviction ratio λ predicts cross-sectional cryptocurrency returns in a manner that differs systematically, and informatively, across asset types. For proof-of-stake coins, where the medium-of-exchange function gives valuation a well-defined fundamental anchor, conviction predicts returns only where valuation has not yet absorbed it, and the high-conviction–high-valuation corner earns sharply negative returns: the conditioning mechanism of the theory, not the level of the signal, is the empirical content. For governance tokens, where conviction is a costly governance signal with no transactional counterpart, the premium is one-dimensional and lives in the extremes of the conviction distribution, is distinct from the strong token reversal effect, and resides largely between protocol categories. A coarse quadrant portfolio does not aggregate these effects into significant alpha at our universe breadth—a power limitation we report in full rather than obscure.

The results carry two broader implications. First, on-chain observability changes what is testable in monetary economics: the locking ratio is, to our knowledge, the first supply-side monetary aggregate measurable at the asset level at monthly frequency without survey or estimation error, and its interaction with valuation behaves as partial-revelation theory predicts in precisely the asset class where valuation is well-anchored. Second, the cross-track asymmetry suggests that markets price observable commitment differently depending on whether a fundamental denominator exists to discipline the pricing: where it does (coins), visible conviction can be over-priced; where it does not (tokens), conviction is slowly and incompletely absorbed. Longer samples, broader staking-data coverage, and fee-based token fundamentals—each currently constrained by data infrastructure rather than concept—are the natural extensions.

A Derivation of the SoV/MoE Decomposition

This appendix provides the formal derivation of the SoV/MoE decomposition and its simplification to a function of λ alone.

Setup. Let M denote the total circulating supply of a digital asset and MC its market capitalization, which serves as the money supply proxy in the QTM framework. Let PQ denote nominal economic output (on-chain transaction volume for coins; protocol throughput for tokens). Standard QTM gives $\mathrm{MC} \cdot V = PQ$, so the market capitalization satisfies $\mathrm{MC} = PQ/V$.

The locking ratio. Define λ as the proportion of total supply that is locked or staked:

$$\lambda = \frac{M_{\text{locked}}}{M}, \qquad \lambda \in [0, 1).$$

The free-circulating supply is $M(1-\lambda)$. We define the *free velocity* V_{free} as the velocity of only the economically active (unlocked) portion:

$$V_{\text{free}} = \frac{PQ}{M(1-\lambda)}.$$

MoE and SoV components. The Medium of Exchange component of market capitalization is the value attributable to current transactional activity, priced at the free velocity:

$$\text{MC}_{\text{MoE}} = \frac{PQ}{V_{\text{free}}} = M(1-\lambda).$$

The Store of Value component is the residual:

$$\text{MC}_{\text{SoV}} = \text{MC} - \text{MC}_{\text{MoE}} = M - M(1-\lambda) = M\lambda.$$

The SoV/MoE ratio. The ratio of the SoV to MoE components is:

$$\frac{\text{MC}_{\text{SoV}}}{\text{MC}_{\text{MoE}}} = \frac{M\lambda}{M(1-\lambda)} = \frac{\lambda}{1-\lambda}.$$

This result shows that the SoV/MoE ratio is a monotone increasing function of λ alone, requiring no estimation of velocity, transaction volume, or any other derived quantity. As $\lambda \to 0$, the ratio approaches zero (pure medium of exchange). As $\lambda \to 1$, the ratio diverges (pure store of value).

The Growth-Levelized NVT ratio. The Network Value to Transactions ratio is NVT = $\text{MC}/PQ_{\text{annualized}}$. We construct a forward-looking version by replacing current PQ with PQ^*, the levelized present value of the projected transaction stream:

$$PQ^* = \frac{\sum_{s=1}^{n} \frac{PQ_0(1+g)^s}{(1+r_e)^s} + \frac{PQ_0(1+g)^n(1+g_\infty)}{(r_e-g_\infty)(1+r_e)^n}}{\text{annuity factor}(r_e, n)},$$

where g is the trailing k-year CAGR of PQ, r_e is the CAPM-estimated required return, g_∞ is the terminal growth rate, and the annuity factor converts the present value to a levelized annual equivalent (analogous to the Excel PMT function). The Growth-Levelized NVT ratio is then $\text{NVT}_{GL} = \text{MC}/PQ^*$. Full estimation details are in Section 4.

B Additional Tables

This appendix collects the data-panel coverage table, the signal-by-signal horse-race results, and the full interaction specifications under alternative valuation ratios.

[Table 13 about here.]

[Table 14 about here.]

[Table 15 about here.]

C Sample Composition

Tables 16 and 17 list the assets in the two regression samples. Months counts the asset-months each contributes to its sample; membership requires simultaneous observation of the conviction and valuation measures, so an asset's window reflects its data coverage, not its listing history.

[Table 16 about here.]

[Table 17 about here.]

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Table 1: Sample Construction

	Excluded	Remaining
CoinMarketCap universe, 2015-08 to 2026-05		1,939
Less: other class (stablecoins, wrapped, infrastructure)	858	1,081
Coin track		
Coins		633
Less: proof-of-work only (no staking channel)	323	310
Less: consensus type unverifiable	24	286
Less: historical staking data unrecoverable	256	30
Less: no on-chain volume overlap (NVT_GL)	6	**24**
Token track		
Tokens		448
Less: no DeFiLlama TVL coverage	299	149
Less: no conviction channel observed	3	146
Less: TVL history insufficient for growth estimate	45	**101**
Combined regression panel		**125**

Notes: Each row removes assets failing the stated requirement, conditional on surviving all prior rows. The coin regression sample requires proof-of-stake architecture, at least one month of reconstructed staking-based λ_z, and overlapping positive NVT_GL. The token regression sample requires DeFiLlama TVL coverage, at least one conviction channel, and sufficient TVL history for the growth-levelized denominator TVL* (at least one year).

Table 2: Summary Statistics of Main Variables

	N	Mean	SD	p10	p25	Median	p75	p90
Panel A: Coin regression sample (24 assets, 2020-12–2026-05)								
Staking share λ (%)	671	40.8	22.2	13.3	21.8	38.6	57.0	71.4
SoV/MoE ratio $\lambda/(1-\lambda)$	671	5.98	66.8	0.15	0.28	0.63	1.32	2.50
NVT_GL	718	1.60	6.23	0.00	0.00	0.02	0.50	2.65
Market cap ($M)	718	36,994	92,232	74.6	295.1	823.4	12,950	112,831
Monthly return t+1 (%)	694	−1.02	28.9	−27.8	−17.4	−4.42	7.64	28.2
Beta (36m, vs. BTC)	718	1.18	0.42	0.68	0.96	1.12	1.37	1.71
Panel B: Token regression sample (101 assets, 2021-01–2026-05)								
Conviction index λ_z	2,771	−0.05	0.73	−0.86	−0.60	−0.18	0.38	1.03
NV/TVL_GL	2,771	411.4	2,503	0.00	0.20	2.29	16.1	281.2
TVL ($M)	2,771	1,017	3,433	0.63	5.48	42.2	425.2	2,351
Price-to-fees (P/F)	1,367	190.9	1,506	1.97	4.32	12.1	33.8	127.1
Market cap ($M)	2,771	370.0	908.7	20.1	40.4	97.7	303.2	783.7
Monthly return t+1 (%)	2,658	−1.49	31.5	−29.7	−19.3	−7.57	8.49	32.4
Beta (36m, vs. BTC)	2,771	1.19	0.50	0.61	0.91	1.16	1.45	1.80

Notes: Unwinsorized distributions over the regression samples. The staking share and SoV/MoE rows cover the 671 coin-months with a direct staking observation (the remaining 47 use the composite-index fallback and have no ratio representation). P/F covers the 1,367 fee-covered token-months. Heavy right tails in NVT_GL, NV/TVL_GL, TVL, P/F, and market capitalization motivate the log transforms and 1%/99% winsorization applied before estimation. Median forward returns are negative in both panels: the sample is dominated by the post-2021 contraction, which cross-sectional (within-month) tests are immune to but portfolio long legs are not. Coverage counts for the underlying data panels appear in Appendix Table 13.

Table 3: Coins: Conviction, Valuation, and Their Interaction

	(1)	(2)	(3)	(4)	(5) Cheap	(6) Expens.	(7) Diff.
Conviction	0.0055	−0.0019	−0.0022	−0.0100*	0.0070	−0.0303	0.0098
	(0.69)	(−0.25)	(−0.28)	(−1.67)	(1.15)	—	(1.38)
Valuation			0.0059	0.0043			
			(0.94)	(0.84)			
Conviction × Valuation				−0.0169***			
				(−3.54)			
Conviction × High-val							−0.0420***
							(−3.99)
Controls	No	Yes	Yes	Yes	Yes	Yes	Yes
Observations	686	680	680	680	356	324	680

Notes: Conviction is the log-odds staking share $\ln(\lambda/(1-\lambda))$; valuation is $\ln \mathrm{NVT\_GL}$; both standardized within class-month. Controls: log market cap, momentum (3m, 12–2), lagged return, 36-month beta. Columns (5)–(6) split at the class-month median valuation; column (7) tests the difference. The column-(6) t-statistic is suppressed: with 24 asset clusters the split-sample statistic (−10.9) is unreliable, and inference rests on the column-(7) difference test.

Table 4: Tokens: Governance Conviction and Returns

	(1)	(2)	(3)	(4)	(5) FM	(6) FM
Conviction	0.0064**	0.0056	0.0054	0.0060	0.0070*	0.0080**
	(1.97)	(1.51)	(1.49)	(1.52)	(1.77)	(2.19)
Valuation				0.0062		0.0014
				(1.60)		(0.20)
Conviction × Valuation				0.0035		0.0024
				(0.78)		(0.56)
Return (t)		−0.0144**	−0.0151**	−0.0156**	−0.0213***	−0.0178***
		(−2.21)	(−2.33)	(−2.45)	(−3.39)	(−3.96)
Sector FE (coarse)	No	No	Yes	Yes	—	—
Controls	No	Yes	Yes	Yes	Yes	Yes
Observations	2,652	2,592	2,592	2,592	52 mo	50 mo

Notes: Conviction is the composite index λ_z; valuation is $\ln \mathrm{NV/TVL\_GL}$. Coarse sectors: DEX, Lending, Yield, Derivatives, Staking/LSD, Other. Columns (5)–(6) are Fama–MacBeth monthly cross-sections with Newey–West (3 lags) t-statistics. The strong negative coefficient on Return(t) is the token short-horizon reversal, a notable effect in its own right and the benchmark the conviction signal must beat in Section 5.5.

Table 5: Stars-minus-Avoid Quadrant Portfolios

Portfolio	Months	α/mo	t	β_{CMKT}	R^2	Sharpe
Coin EW	20	−0.0338	(−1.54)	0.10	0.26	−0.47
Coin VW	20	0.0535	(1.39)	0.12	0.02	0.75
Token EW	49	0.0119	(0.84)	−0.22	0.15	0.36
Token VW	49	0.0276	(1.13)	−0.46	0.24	0.52
EW universe benchmark	129					0.61

Notes: Median-split quadrants within class-month; ≥ 4 assets per leg required; monthly rebalancing. Alphas vs. monthly CMKT/CSMB/CMOM analogs of Liu, Tsyvinski, and Wu (2022) constructed from the 1,939-asset universe, Newey–West (3 lags). The monthly momentum factor analog carries a negative own-premium (−4.6%/month, $t = -1.57$), a documented consequence of monthly aggregation; factor construction details in Section 4.

Table 6: Token Conviction-Only Long-Short Portfolios

Sort	Mean/mo	α/mo	t	Sharpe	Months	Post-23 α (t)
Median EW	0.0032	0.0033	(0.38)	0.14	55	−0.0114 (−1.82)
Tercile EW	0.0080	0.0036	(0.37)	0.21	50	−0.0068 (−0.88)
Quintile EW	**0.0228**	**0.0171****	**(2.18)**	**0.86**	**50**	**0.0148* (1.79)**
net 25 bps		0.0162**	(2.06)		50	
net 50 bps		0.0153*	(1.95)		50	
Quintile VW	0.0369	0.0311*	(1.83)	0.92	50	0.0217 (1.38)
Decile EW	0.0075	0.0003	(0.02)	0.14	46	0.0116 (0.71)
Quintile EW, sector-neutral	0.0088	0.0040	(0.39)	0.24	50	−0.0075 (−0.89)

Notes: Long top / short bottom conviction bucket within token class-month, formation on information through t, held for month t+1; minimum 3 names per leg. Alphas vs. monthly LTW-analog factors, Newey–West (3 lags). Sector-neutral: conviction demeaned within coarse sector-month before sorting. Leg turnover for the quintile EW is 0.16–0.19 per month, so the 50 bps haircut costs ∼18 bps/month.

Table 7: Competitor Portfolios and Spanning Tests (Tokens)

	Competitor's own α		Conviction q5 α, competitor added	
Competitor long-short	α/mo	t	α/mo	t
None (baseline)	—		0.0171**	(2.18)
Reversal	−0.0468***	(−3.02)	0.0253***	(3.18)
Momentum (3m)	0.0135	(1.21)	0.0153*	(1.85)
Momentum (12–2)	−0.0060	(−0.37)	—	
52-week high	−0.0217	(−1.26)	0.0203***	(2.70)
Size	−0.0035	(−0.23)	0.0170**	(2.15)
MA-distance	−0.0217	(−1.57)	0.0183**	(2.46)
Volatility	−0.0106	(−0.73)	0.0169**	(2.10)
Idiosyncratic volatility	−0.0056	(−0.43)	0.0169**	(2.15)
Amihud illiquidity	−0.0147	(−0.82)	0.0174**	(2.14)
Skewness	0.0031	(0.16)	0.0166**	(2.11)
P/F (cheap − expensive)	−0.0018	(−0.10)	0.0122	(1.67)
Revenue DCF (cheap − expensive)	0.0098	(0.55)	0.0154*	(1.84)
Momentum family + size (four jointly)			0.0234**	(2.60)
Completed battery (twelve jointly)			0.0174**	(2.45)
Completed battery + fee comparators			0.0217***	(2.82)

Notes: All portfolios are EW quintile long-shorts built identically within token class-month (min. 3 names per leg), evaluated against the monthly LTW factors with Newey–West (3 lags). Left columns: does the competitor signal itself earn alpha? (None does; reversal is a significantly *negative*-alpha strategy.) Right columns: does the conviction quintile's alpha survive when the competitor is added as a factor? The continuous MA-distance signal supersedes a binary MA-crossover whose legs were populated in only 28–31 months. The P/F and revenue-DCF cells run on fee-covered subsamples (41 and 38 months); their lower alphas reflect sample composition, with small, insignificant competitor loadings. In the reverse direction, no competitor earns positive alpha on the LTW factors plus the conviction quintile.

Table 8: The Horse Race: Joint Multivariate Specifications

	Coins	Tokens (fee-covered)
Conviction	−0.0094	0.0011
	(−1.34)	(0.19)
Conviction × Valuation	−0.0235***	
	(−2.68)	
Valuation (NVT_GL)	0.0003	
	(0.03)	
Price-to-fees (ln P/F)		−0.0090**
		(−2.32)
Raw NVT / raw NV/TVL	0.0207	−0.0035
	(1.33)	(−0.36)
Amihud illiquidity	−0.0276***	−0.0004
	(−4.49)	(−0.08)
Idiosyncratic volatility	−0.0220	−0.0255**
	(−1.31)	(−2.22)
Momentum (12–2)	−0.0038	0.0177***
	(−0.45)	(2.87)
Reversal (return t)	−0.0094	−0.0041
	(−0.86)	(−0.42)
52-week high	−0.0118	−0.0166
	(−0.54)	(−1.34)
Other comparators	included, all $\|t\| < 1.35$	
Observations	644	1,145
Assets	22	47
R^2 (within)	0.032	0.023

Notes: Each column regresses the winsorized one-month-ahead return on all comparator signals simultaneously (stock-to-flow, supply growth, momentum 3m, MA-distance, volatility, and skewness are included but suppressed for space; none exceeds $|t| = 1.35$), with month fixed effects and SEs two-way clustered by asset and month. Month fixed effects absorb all common factor realizations, so coefficients measure cross-sectional predictive content net of the common component. The token column runs on the fee-covered subsample so that the P/F comparator can enter; the conviction slope on that subsample is +0.0014 (t = 0.24) before P/F enters. Signal-by-signal (univariate) results for both tracks appear in Appendix Table 14.

Table 9: Measurement Robustness of the Conviction–Valuation Interaction

Variant	Coin conv × val (t)	Token conv × val (t)
Baseline (growth-levelized, MRP 30%)	−0.0169*** (−3.54)	0.0034 (0.79)
Raw NVT / raw NV/TVL	−0.0060* (−1.90)	0.0002 (0.05)
Exclude growth-cap-binding months	−0.0163 (−0.97)	−0.0024 (−0.54)
Exclude custodial-flagged (B4) months	—	0.0023 (0.55)
Exclude conviction-fallback coin-months	−0.0181*** (−4.67)	—
MRP 20%	−0.0175*** (−3.55)	0.0033 (0.78)
MRP 40%	−0.0163*** (−3.54)	0.0035 (0.80)
Fee-anchored valuation (ln P/F)	—	−0.0024 (−0.45)
Revenue-DCF valuation (ln MC/REV*)	—	−0.0035 (−0.31)

Notes: Each row re-estimates the column-(4) specification of Tables 3–4 under the stated variant. B4 flags apply only to the holding channel and hence only to tokens; the conviction fallback applies only to coins. The fee-anchored rows run on the fee-covered subsamples (49 and 33 tokens); coverage-matched NV/TVL_GL baselines on the identical subsamples are equally insignificant ($t = 0.70$ and -0.80), so the sign flip is not attributable to the fee denominator.

Table 10: Mechanism Discrimination Tests

	Test	Result	Verdict
M1 attention	token conv $\times$ turnover	$+0.0069$ (1.21), wrong sign	not supported
M2 denominator	sector-demeaned valuation	interaction stays + (1.27)	rejected
M3 measurement	raw ratio, cap-excluded	≈ 0 in both	rejected
M4 seigniorage	staking-yield controls	coin int. -0.0188^{***} (-4.39)	coin result survives

Table 11: Token Conviction by Channel

Channel	Tokens	Obs.	Panel slope (t)	Quintile α/mo (t)	Months
Holding duration (HODL-6m)	88	2,468	0.0055 (1.50)	0.0148* (1.71)	50
Protocol staking	13	459	0.0008 (0.09)	too thin to sort	
Governance delegation	9	131	−0.0113 (−0.94)	too thin to sort	
Governance voting	32	580	0.0006 (0.09)	too thin to sort	
Composite λ_z (reference)	97	2,592	0.0056 (1.57)	0.0171** (2.18)	50

Notes: Each row standardizes the raw channel within token-month and re-estimates the baseline panel specification (controls, month FE, two-way clustered SEs) and, where breadth permits, the EW quintile long-short against the LTW factors (NW-3). Channel coverage is determined by protocol design and data availability (Section 3.1). This decomposition was run at the review stage rather than pre-specified.

Table 12: Summary of Findings

Test	Verdict	Key evidence
Hypotheses		
H1 coins: conviction level	Rejected at t+1	t = 0.69 alone; marginally positive at t+6 (t = 1.90)
H1 tokens: conviction level	Qualified support	+0.6–0.8%/mo per SD; significant only alone (t = 1.97) and in Fama–MacBeth (t = 1.77–2.19); $t \approx 1.5$ with controls
H2: coins	Strongly supported	interaction $t = -3.54$; +0.7%/mo (cheap) vs. −3.0%/mo (expensive); survives 12-comparator battery ($t = -2.68$), yield controls ($t = -4.39$), MRP variants; caveats: raw-NVT attenuation, g-cap subsample
H2: tokens	Rejected	interaction $\approx$ 0 across six conditioning variables, ending with fee/revenue anchoring; a fact about tokens, not measurement
H3: quadrant portfolio	Null (power-limited)	no variant significant; breadth guard binds; equally null under P/F and revenue-DCF valuation axes
One-dimensional extension (tokens)		
Conviction quintile long-short	Supported (disclosed as exploratory-then-confirmed)	+1.71%/mo α (t = 2.18); net of 50 bps +1.53% (t = 1.95); post-2023 +1.48% (t = 1.79)
Spanning vs. completed battery	Survives	+1.74% (t = 2.45) vs. twelve long-shorts; +2.53% (t = 3.18) controlling reversal alone
Structure of the premium	In the extremes; present within the deepest category	decile and sector-neutral sorts die (dilution/thin-category noise); DEX-only quintile +3.4%/mo (t = 2.04); no absorption proxy conditions it
Horse race		
Coin comparators	None displace the interaction	Amihud strongest single ($t = -2.65$); 52-wk high strongest post-2024
Token comparators	Reversal dominant; P/F the one working valuation ratio	reversal $t = -2.4$ to -2.6; P/F $t = -2.50$ singles, −2.32 joint; its DCF transform and portfolio versions are dead
Scarcity and technical narratives	Largely unsupported	S2F null everywhere; MA/vol/skew long-shorts all null for tokens

Table 13: Data Panel Coverage

Panel / Sample	Assets	Asset-months	Date range	Median MC
Full universe	1,939	156,838	2015-08 to 2026-05	$6.5M
Conviction panel (λ_z)	467	13,626	2016-12 to 2026-05	—
ch1 staking	48	2,222	2017-07 to 2026-05	—
ch2 holding	408	11,654	2015-08 to 2026-05	—
ch3 delegation	26	627	2020-05 to 2026-05	—
ch3 voting	53	1,202	2020-08 to 2026-05	—
NVT_GL panel	67	2,526	2016-08 to 2026-05	—
TVL panel (raw)	162	8,066	2017-09 to 2026-05	—
NV/TVL_GL panel	111	3,125	2021-01 to 2026-05	—
Coin regression (NVT_GL)	24	718	2020-12 to 2026-05	$820M
Token regression (NV/TVL_GL)	101	2,771	2021-01 to 2026-05	$98M
Combined	125	3,489	2020-12 to 2026-05	$138M

Notes: Median MC is the cross-sectional median market capitalization across all asset-months with a positive price observation. Conviction panel sub-channel counts sum to more than the panel total because 1,725 asset-months appear in two or more channels. Coin regression requires simultaneous non-missing λ_z and positive NVT_GL for proof-of-stake coins. Token regression requires simultaneous non-missing λ_z and positive NV/TVL_GL. Median NVT_GL in the coin regression sample is 0.017. Median NV/TVL_GL in the token regression sample is 2.29 (IQR: 0.20–16.10); median TVL is $42.2M.

Table 14: Signal-by-Signal Horse Race (Univariate)

	Coins		Tokens	
Signal (own coefficient, singles)	Full	2024+	Full	2024+
Conviction	0.0055	0.0090	0.0064**	0.0064*
	(0.69)	(1.22)	(1.97)	(1.88)
Raw NVT / raw NV/TVL	0.0080	0.0162*	−0.0061	−0.0051
	(0.94)	(1.68)	(−1.53)	(−1.05)
Price-to-fees (ln P/F)	—	—	−0.0103**	−0.0148***
			(−2.50)	(−2.70)
Revenue DCF (ln MC/REV*)	—	—	−0.0019	0.0011
			(−0.42)	(0.17)
Stock-to-flow	0.0023	0.0090*	0.0045	0.0012
	(0.28)	(1.86)	(1.03)	(0.28)
Supply growth (12m)	−0.0075	−0.0143***	0.0005	0.0008
	(−0.94)	(−2.79)	(0.14)	(0.14)
52-week high	0.0156	0.0236**	−0.0134**	−0.0047
	(1.52)	(2.06)	(−2.04)	(−0.67)
MA-distance	0.0119	0.0149	−0.0120**	−0.0115
	(1.32)	(1.36)	(−2.21)	(−1.61)
Volatility (12m)	−0.0120	−0.0193**	0.0000	−0.0039
	(−1.44)	(−2.29)	(0.00)	(−0.79)
Idiosyncratic volatility	−0.0146*	−0.0209**	−0.0046	−0.0098***
	(−1.67)	(−2.10)	(−1.39)	(−2.62)
Amihud illiquidity	−0.0223***	−0.0286***	0.0043	0.0036
	(−2.65)	(−3.23)	(0.85)	(0.71)
Return skewness (36m)	−0.0099	−0.0125	−0.0040	−0.0076
	(−1.15)	(−1.13)	(−1.04)	(−1.49)
Reversal (return t)	0.0019	−0.0026	−0.0142**	−0.0195***
	(0.21)	(−0.25)	(−2.39)	(−2.62)
Momentum (3m)	0.0081	0.0082	−0.0019	−0.0017
	(1.00)	(0.87)	(−0.38)	(−0.28)
Momentum (12–2)	0.0093	0.0190**	−0.0021	0.0069
	(0.87)	(2.10)	(−0.46)	(1.47)
Joint (all comparators simultaneously):				
Conviction × Valuation (coins)	−0.0235***	−0.0188***		
	(−2.68)	(−3.17)		
Conviction (tokens)			0.0011	
			(0.19)	
ln P/F (tokens)			−0.0090**	
			(−2.32)	

Notes: Each "singles" cell is the own coefficient from a regression of the winsorized one-month-ahead return on the standardized signal with month fixed effects, SEs two-way clustered. Fee-based rows run on the fee-covered subsamples (49–55 of 101 tokens, histories mostly 2021+). The joint token specification is estimated on the fee-covered subsample; the token conviction slope on that subsample is +0.0014 ($t = 0.24$) before the P/F column enters — the fee multiple neither spans nor revives it.

Table 15: Hypothesis 2 for Tokens under Alternative Valuation Ratios

Valuation ratio used	NV/TVL_GL (baseline)	ln P/F (fee multiple)	ln MC/REV* (revenue DCF)
Conviction	0.0061	0.0066	−0.0009
	(1.47)	(1.15)	(−0.11)
Valuation (level)	0.0042	−0.0119***	−0.0014
	(0.79)	(−3.21)	(−0.25)
Conviction × Valuation	0.0034	−0.0024	−0.0035
	(0.79)	(−0.45)	(−0.31)
Conviction, cheap half	0.0017	0.0085	0.0050
	(0.28)	(0.94)	(0.37)
Conviction, expensive half	0.0041	0.0023	−0.0096
	(1.13)	(0.37)	(−0.65)
Split difference (conv × high)	0.0048	−0.0068	−0.0184
	(0.65)	(−0.89)	(−0.87)
Coverage-matched baseline interaction	—	0.0046	−0.0102
		(0.70)	(−0.80)
Observations / tokens	2,592 / 97	1,282 / 49	915 / 33

Notes: Each column estimates the full interaction specification (conviction, controls, valuation, conviction × valuation; month FE; two-way clustered SEs) with the stated valuation ratio. The coverage-matched baseline row re-estimates the NV/TVL_GL interaction on the identical fee- or revenue-covered subsample: it is equally insignificant, so the sign change under fee anchoring is a sample-composition effect, not a denominator effect. H2 predicts a negative interaction; under the fee measures the signs finally comply but no statistic approaches significance. The significant P/F *level* coefficient is the horse-race finding of Table 8, not an H2 result.

Table 16: Coin Regression Sample (24 assets)

Symbol	Name	Months	Median MC ($M)
ETH	Ethereum	66	284,715
BNB	BNB	23	92,775
SOL	Solana	40	63,535
ADA	Cardano	43	13,896
TRX	TRON	58	9,651
AVAX	Avalanche	51	7,763
CRO	Cronos	11	3,496
FLR	Flare	14	1,148
POL	Polygon (prev. MATIC)	9	1,096
XDC	XDC Network	36	736
EGLD	MultiversX	32	625
KLAY	Klaytn	20	604
CORE	Core	27	519
STRK	Starknet	16	426
RON	Ronin	28	412
KAVA	Kava	26	410
CHZ	Chiliz	9	404
GNO	Gnosis	47	387
CELO	Celo	51	289
GLMR	Moonbeam	40	160
S	Sonic	5	118
PEAQ	peaq	1	67
MOVR	Moonriver	45	57
XRD	Radix	20	53

Table 17: Token Regression Sample (101 assets)

Symbol (coarse sector, months), ordered by median market capitalization		
UNI (DEX, 56)	ONDO (Derivatives, 17)	ARB (Other, 27)
ENA (Other, 13)	SKY (DEX, 1)	AAVE (Lending, 49)
LDO (LSD, 39)	FLOKI (Other, 22)	MORPHO (Lending, 6)
CRV (DEX, 57)	SNX (DEX, 65)	CAKE (DEX, 39)
PENDLE (Yield, 24)	AERO (DEX, 14)	COMP (Lending, 59)
AXL (Other, 27)	1INCH (DEX, 47)	ETHDYDX (Derivatives, 37)
SUPER (Other, 37)	CVX (Yield, 44)	W (Other, 14)
LRC (DEX, 53)	FRAX (DEX, 42)	WOO (DEX, 32)
SYRUP (Lending, 2)	YFI (DEX, 58)	AMP (Other, 50)
T (Lending, 38)	GMX (DEX, 32)	SUSHI (DEX, 52)
BAL (DEX, 53)	BICO (Other, 31)	EIGEN (LSD, 8)
SXP (DEX, 43)	ILV (Other, 5)	APE (Yield, 3)
VVS (DEX, 38)	KEEP (Other, 46)	METIS (Other, 37)
XVS (Lending, 29)	KNC (DEX, 47)	CELR (Other, 37)
NMR (Derivatives, 46)	REP (Other, 31)	STG (Other, 28)
BONE (DEX, 30)	SPELL (Lending, 42)	LQTY (Lending, 28)
BNT (DEX, 50)	REN (Other, 31)	HFT (DEX, 27)
RBN (Lending, 18)	LON (DEX, 42)	SYN (Yield, 27)
MDX (DEX, 29)	AURORA (Yield, 28)	CORE (Yield, 13)
BADGER (Yield, 42)	AEVO (DEX, 15)	TRU (Lending, 38)
AUCTION (Other, 17)	GNS (Derivatives, 26)	LINA (DEX, 22)
DODO (DEX, 49)	STRK (Lending, 30)	BAKE (DEX, 33)
ORN (DEX, 3)	PERP (DEX, 34)	MLN (DEX, 41)
MAV (DEX, 19)	IDEX (DEX, 52)	ALPHA (Lending, 8)
BLAST (Yield, 11)	WXT (Lending, 35)	TT (DEX, 19)
OSMO (DEX, 2)	BZRX (DEX, 3)	YFII (Yield, 26)
EPIC (Other, 14)	ALCX (Yield, 37)	FARM (Yield, 44)
KP3R (Derivatives, 17)	CPOOL (Lending, 6)	SWAP (DEX, 10)
EPS (DEX, 37)	ANC (Lending, 6)	STAKE (Other, 12)
MET (DEX, 11)	TIME (Lending, 19)	PNT (Other, 7)
HEGIC (DEX, 43)	ICHI (Yield, 18)	ZKB (DEX, 2)
MIR (Derivatives, 4)	NFTX (Other, 21)	TLM (Other, 1)
PEAK (Yield, 6)	DDX (DEX, 13)	RGT (Yield, 13)
BTCST (Yield, 4)	NVT (DEX, 1)	

Notes: Coarse sectors follow the mapping of Section 6: DEX, Lending, Yield, Derivatives, LSD (liquid staking), and Other.